Chapter 14: Semiconductor Electronics

Formula Sheet + Tips & Tricks

Formula Sheet

I. p-n Junction Diode – Forward Bias

Formula: $V_{\text{battery}} = I_{\max}R + V_D$

V_{battery} = battery EMF, $I_{\max}$ = maximum safe forward current, R = series resistance, V_D = constant forward voltage drop across the diode.

Formula: $R_{\text{dynamic}} = \dfrac{\Delta V}{\Delta I}$

R_{dynamic} = dynamic (AC) resistance of the diode, ΔV = change in voltage across diode, ΔI = corresponding change in diode current.

Formula: $I = \dfrac{V - V_0}{R}$

I = current in the circuit, V = applied battery voltage, V_0 = barrier (junction) potential of the diode, R = series resistance.

Tips, Tricks & Hacks

- In forward bias, the depletion layer narrows and resistance drops (diode conducts easily); in reverse bias, it widens and resistance rises (diode blocks, except for a tiny leakage current) — state cause (depletion width change) before effect (current) for full marks.
- Diode symbol arrow points in the direction of *conventional current flow* when forward biased (p to n externally, i.e. from p-side terminal, through diode, to n-side) — a quick way to read circuit diagrams instantly.
- Zener diode is operated deliberately in *reverse breakdown* (unlike a normal diode, which you avoid pushing into breakdown) — this is precisely what makes it useful

as a voltage regulator.

- A full-wave rectifier (using two diodes or a bridge of four) gives twice the output frequency and higher average output than a half-wave rectifier — know this comparison, it's a classic short-answer/diagram question.
- For a p-n junction, the barrier potential is typically $\sim 0.7\,\mathrm{V}$ for silicon and $\sim 0.3\,\mathrm{V}$ for germanium — if a question doesn't specify, silicon values are the safer default assumption in Indian board exams.
- Photodiode is operated in reverse bias (so that a small change in reverse current due to light can be measured clearly against a low, stable background current) — remembering "reverse bias for sensing, forward bias for conducting" separates diode-as-sensor from diode-as-rectifier questions.